

A new Source-Pathway-Receptor based model for identification and evaluation of potentially contaminated sites

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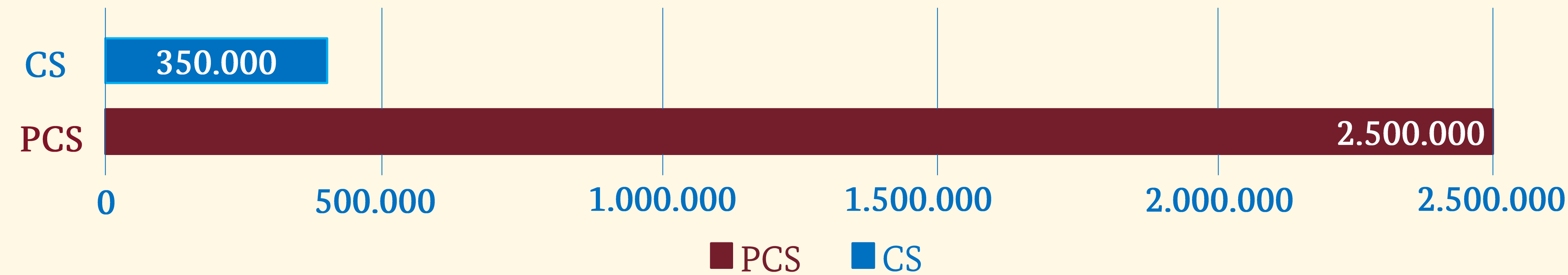
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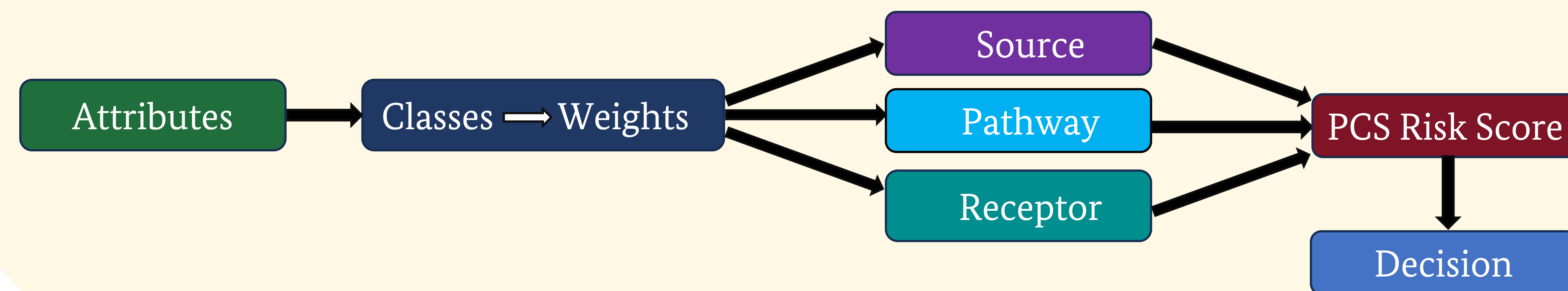
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1 Motivation and Background

- Numerous Contaminated (CS) and Potentially Contaminated (PCS) around the world
- Risk assessment based on Multicriteria Decision Analysis (MCDA)
- Criteria: Variety of attributes



SPR paradigm to analyze PCS



3 Results

Method		Plume length [m]	Plume area [m ²]	Plume width [m]
SPR _{Model}	r_s	0.83	0.90	0.75
	P -value	Strong correlation $r_s > \text{Critical value}$ 6.93×10^{-6} reject H_0	Very strong correlation $r_s > \text{Critical value}$ 5.98×10^{-8} reject H_0	Strong correlation $r_s > \text{Critical value}$ 1.47×10^{-4} reject H_0
Regional Risk Assessment	r_s	0.53	0.76	0.68
	P -value	Moderate correlation $r_s > \text{Critical value}$ 1.67×10^{-2} reject H_0	Strong correlation $r_s > \text{Critical value}$ 1.04×10^{-4} reject H_0	Moderate correlation $r_s > \text{Critical value}$ 5.53×10^{-3} reject H_0
Hazard Risk Score	r_s	-0.03	0.09	0.129
	P -value	Very weak correlation $r_s < \text{Critical value}$ 0.915 Do not reject H_0	Very weak correlation $r_s < \text{Critical value}$ 0.719 Do not reject H_0	Very weak correlation $r_s < \text{Critical value}$ 0.587 Do not reject H_0
Prioritization of Oil Contaminated Sites	r_s	-0.33	-0.33	-0.02
	P -value	A weak correlation $r_s < \text{Critical value}$ 0.152 Do not reject H_0	A weak correlation $r_s < \text{Critical value}$ 0.151 Do not reject H_0	Very weak correlation $r_s < \text{Critical value}$ 0.931 Do not reject H_0

Significance level $\alpha = 0.05$ Critical value = 0.446 H_0 : There is no monotonic association between the two variables

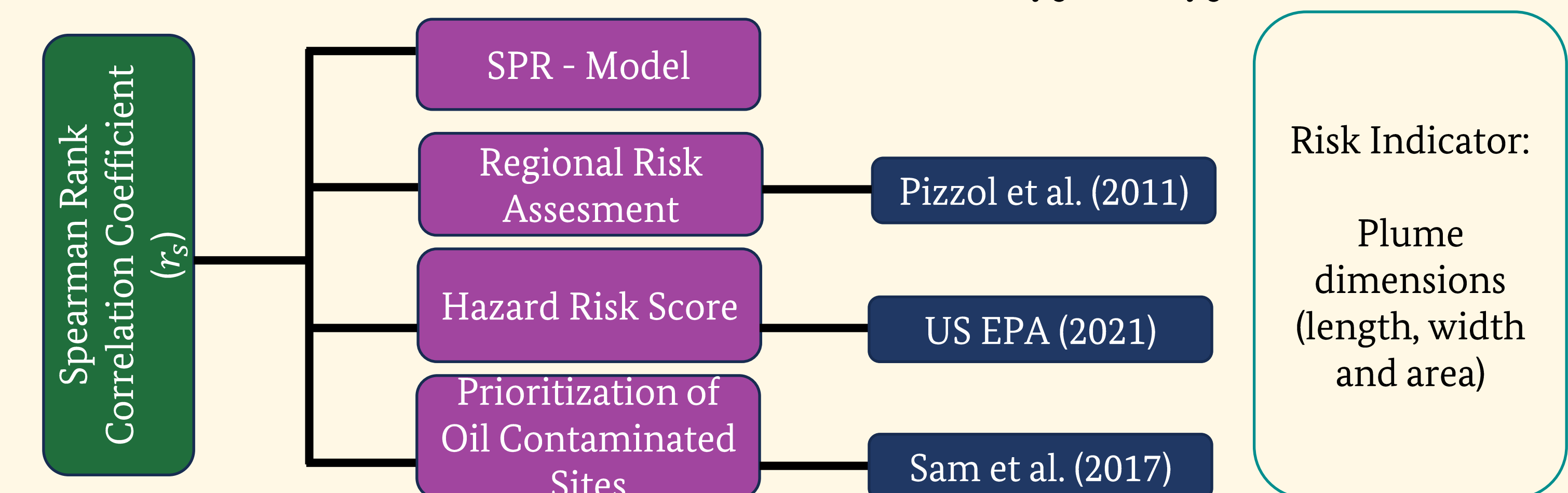
2 Methods

- Dataset: Total of 20 contaminated sites of steady-state plumes
- For interaction among attributes (Pathway and Receptor): Choquet Integral (Choquet, 1954)

$$\int x d\mu = \sum_{i=1}^n (x^{(i)} - x^{(i-1)}) \mu(\{(i) \dots (n)\})$$

Spearman Rank Correlation Coefficient (r_s):
$$r_s = 1 - \frac{6 \sum_{i=1}^n d^2}{n^3 - n}$$

Three other methods for risks in terms of scoring and attributes in SPR scheme



- Simple model parameters and attributes (allocated in Source and Pathway):

Parameters	Methods	SPR _{Model}	Regional Risk Assessment	Hazard Risk Score	Prioritization of Oil Contaminated Sites
Model domain		Contamination origin size	Size of the site	-	-
Time		Years of operation	Years of operation	-	-
Hydraulic conductivity		Overall lithology (texture)	Transmissivity	-	-
Porosity		Homogeneous or heterogeneous porous media	-	-	-
Partner reactant		Hydrological stressor (Recharge)	-	-	-
Contaminant concentration		Level of contamination	-	Hazardous waste quantity (Tier A, B, C or D)	Level of contamination
Source geometry		Source geometry	-	Type (AST, UST, Drums, Pile, Landfill, etc.)	-
Reaction		Contamination reactivity, Depth to GW table	Depth to GW table, N.of the water bearing layers	Waste/Substance characteristics	-

4 Conclusions

- SPR_{Model} has a simplified but methodical and coherent structure that relies on conceptual site model
- Selection of key attributes was based on simple transport model parameters
- r_s : strong to very strong correlation between plume dimensions and SPR_{Model} results
- In comparison to other methods, SPR_{Model} shows better results of correlation

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